ADITYA BABAR

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PROFESSIONAL SUMMARY

Final-year Electronics and Computer Science undergraduate (2026 batch) with strong foundations in Data Structures & Algorithms, OOP, DBMS, and Computer Networks. Solved 100+ DSA problems across arrays, strings, trees, and dynamic programming. Delivered production-grade software including an ML pipeline achieving 98% accuracy across 860K+ records and full-stack web applications. Proficient in Python, Java, and C with hands-on experience in REST APIs, Django, Flask, and PostgreSQL.

TECHNICAL SKILLS

Languages: Python, Java, C, SQL

CS Fundamentals: Data Structures & Algorithms (Arrays, Strings, Linked Lists, Trees, DP, Graphs), OOP, DBMS, Computer Networks, Operating Systems

Frameworks & Backend: Django, Flask, REST APIs, Node.js

Frontend: React.js, HTML5, CSS3, Bootstrap, Axios

Databases: PostgreSQL, MySQL, MongoDB, SQLite

Machine Learning / AI: Scikit-learn, NLP, TF-IDF, Logistic Regression

DevOps & Tools: Git, GitHub, Linux, CI/CD Pipelines, Docker (familiar), Waterfall & Agile methodology, unit testing

IoT & Embedded: ESP32, MQTT, C++, Sensors

PROJECTS

ScrapX — Scrap Material Marketplace Platform | React.js · Django REST · PostgreSQL · Razorpay · JWT / *Feb 2026*
github.com/adityababar07/scrapx-backend – *Full-Stack E-Commerce Platform*

- Architected and built a full-stack B2C e-commerce platform end-to-end for scrap material trading (metal, plastic, e-waste, paper) — covering frontend, backend REST APIs, database design, and cloud deployment on github pages and PythonAnywhere.
- Integrated Razorpay payment gateway with secure order lifecycle management; implemented JWT-based role authentication (Djoser) supporting buyer, seller, and admin access levels.
- Engineered a content-based product recommendation engine in Python to surface relevant scrap listings, improving buyer discovery and platform engagement.
- Built a seller analytics dashboard and admin panel providing real-time insights on listings, transactions, and platform activity — demonstrating full product ownership from DB schema to UI.

Phishing URL Detection System | Flask · Python · ML · Scikit-learn / *Sept 2025*
github.com/adityababar07/phishingURLDetectionSystem – *AI-Based Cybersecurity Project*

- Engineered an end-to-end ML pipeline detecting phishing URLs with 98% accuracy across 860K+ records, demonstrating scalable data engineering and model deployment skills valued in Specialist Programmer roles.
- Developed NLP-based TF-IDF character n-gram feature extraction to handle obfuscated, typosquatted, and adversarial URLs at production scale.
- Built scalable data pipelines covering ingestion, preprocessing, model training, and real-time inference; serialized and deployed models via Pickle and Flask REST APIs.
- Optimized Logistic Regression classifier by validating performance with confusion matrix analysis, precision/recall metrics.

Smart Home Automation System | ESP32 · C++ · IoT · MQTT · Android / *May 2025*
github.com/adityababar07/Miniproject2 – *Embedded AI & IoT*

- Architected an ESP32-based home automation system integrating sensors, MQTT communication, and Android remote control, showcasing proficiency in ECE-aligned hardware-software integration.
- Migrated firmware from MicroPython to C++ to enhance execution determinism, reliability, and system performance in a resource-constrained embedded environment.
- Designed and validated full hardware-software integration demonstrating real-time embedded systems design and IoT communication protocols.

ACHIEVEMENTS & COMPETITIONS

- TCS HackQuest Season 10 (National Level) – Competed and cleared Round 1, solving binary exploitation and password cracking challenges.
- Smartcops Hackathon 2025, Sangli – Built a Chrome extension for real-time phishing detection using ML; implemented TF-IDF vectorization and synthetic data augmentation for improved model robustness.
- Nirmiti 2025 – Electro-Mania (National Level) – Secured 2nd Prize across multi-round competition covering General Aptitude, C Programming, and digital logic hardware testing with ICs.

EDUCATION

B.Tech(Electronics and Computer Science) – PVPIT, Budhgaon 2022 – 2026
CGPA: 7.51 / 10.0

HSC – Appasaheb Birnale Public School, Sangli 2021
Percentage: 77.4%

SSC – PM SHRI Kendriya Vidyalaya Painavu, Idukki 2019
Percentage: 95%